Introduction to
Internet of Things
Assignment-Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Based on functionality, MQTT is a _______ protocol.

a. Transport
b. Data
c. Semantic
d. None of these

Correct Answer: b. Data

Detailed Solution: MQTT is a Data Protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 01:30

QUESTION 2:

MQTT is designed for -

a. Remote connections
b. Limited bandwidth
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: c. Both (a) and (b)

Detailed Solution: MQTT is designed for –

1. Remote connections

2. Limited bandwidth

See lecture 6 (Basics of IoT Networking – Part II) @ 03:50

QUESTION 3:

State True or False.

MQTT protocol follows _____________ paradigm for exchanging messages.

1. Client-Server
2. Publish-Subscribe
3. Both (a) and (b)
4. None of these

See lecture 6 (Basics of IoT Networking – Part II) @ 02:00

QUESTION 4:

State True or False.

Statement: “In MQTT, the Subscribers are Lightweight Sensors.”

a. True
b. False

Correct Answer: b. False

Detailed Solution: In MQTT, the Publishers are lightweight sensors.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:49

QUESTION 5:

Which of the following is MQTT component?

a. Middleman
b. Mules
c. Both (a) and (b)
d. None of these

Correct Answer: d. None of these

Detailed Solution:

Components of MQTT are –

1. Publishers
2. Subscribers
3. Beokers

See lecture 6 (Basics of IoT Networking – Part II) @ 04:50

QUESTION 6:

State True or False.

A topic in MQTT can only be numbers.

a. False
b. True

Correct Answer: b. False

Detailed Solution: A topic in MQTT is a string.

Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 7:

State True or False.

There are only two methods specified by the MQTT protocol.

a. False
b. True

Correct Answer: a. False

Detailed Solution: There are 5 number of methods in MQTT protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 05:49

QUESTION 8:

The Publish/Subscribe architecture in MQTT is _____________ driven.

a. Event
b. Pulse
c. Sound
d. None of these

Correct Answer: a. Event

Detailed Solution: Publish/Subscribe in MQTT is event-driven and enables messages to be pushed to clients.

See lecture 6 (Basics of IoT Networking – Part II) @ 08:32

QUESTION 9:

State True or False.

The topic is the routing information for the broker.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The topic is the routing information for the broker.

See lecture 6 (Basics of IoT Networking – Part II) @ 08:30

QUESTION 10:

CoAP is _____________ and ____________.

a. Based on HTTP
b. Is designed for M2M applications
c. None of these
d. Both (a) and (b)

Correct Answer: d. Both (a) and (b)

Detailed Solution: CoAP is based on HTTP and is designed for M2M applications.

See lecture 7 (Basics of IoT Networking – Part III) @ 00:49

QUESTION 11:

In CoAP, client-server interaction is asynchronous over a datagram transport protocol such as _____________.

a. UDP
b. TCP
c. IP
d. XMP

Correct Answer: a. UDP

Detailed Solution: In CoAP, client-server interaction is asynchronous over a datagram transport protocol such as UDP.

See lecture 7 (Basics of IoT Networking – Part III) @ 00:50

QUESTION 12:

What is the full form of AMQP?

a. Advanced Message Querying Protocol
b. Advanced Message Quality Protocol
c. Advanced Message Queuing Protocol
d. None of these

Correct Answer: c. Advanced Message Queuing Protocol

Detailed Solution: Advanced Message Queuing Protocol

See lecture 7 (Basics of IoT Networking – Part IV) @ 0:55

QUESTION 13:

AMQP has _______ number of frame types.

a. 6
b. 3
c. 5
d. 9

Correct Answer: d. 9

Detailed Solution: In AMQP there are nine frame types..

See lecture 8 (Basics of IoT Networking – Part IV) @ 07:20

QUESTION 14:

State True or False.

Statement: "The OSI model has 7 layers."

a. True
b. False

Correct Answer: a. True

Detailed Solution: The OSI model is a conceptual framework that divides any networked communication system into seven layers.

See Page number – 10, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

The “Destination Address” in the IPv4 packet represents which of the following?

a. The source node address of the packet
b. The intermediate hop in the network
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: d. Neither (a) nor (b)

Detailed Solution: The “Destination Address” in the IPv4 packet represents the address of the destination node in the network.

See Page number – 18, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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